Landslides in the Santa Monica Mountains:

A Comprehensive Risk Analysis

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Introduction

The Santa Monica Mountains are a rugged coastal range in Southern California known for their steep terrain, complex geology, and natural beauty. However, they face a prevalent and growing risk: landslides. Landslides are hazardous downslope movements of rock, soil, and debris and are a natural part of mountain processes (Highland & Bobrowsky, 2008). However, such landslides in this area pose a major threat to local communities, infrastructure, and ecosystems. The Santa Monica Mountains' inherent susceptibility arises from its steep slopes, variable geology, and weathered rock formations (Campbell, 1975). This baseline vulnerability is further amplified by natural disturbances, including increasingly frequent and intense wildfires and altered rainfall patterns (Keeley & Syphard, 2016). Given these compounding factors, a comprehensive risk assessment is essential to guide disaster preparedness efforts and inform effective disaster response and planning. *Our study investigates where the most at-risk areas for landslides are in the Santa Monica Mountains by utilizing a hazard-exposure-vulnerability framework, with the goal of providing actionable insights that can help safeguard local communities and ecosystems.*

Methods

I. Study Area

Our study focuses on the Santa Monica Mountains National Recreation Area (SMMNRA), a nationally protected region spanning approximately 156,000 acres. To limit the scope of our analysis, we adjust the formal boundaries of the SMMNRA, stopping at interstates 405 and 101. To conduct this study, we applied a hazard-exposure-vulnerability risk framework. Hazard is defined as: the physical and environmental conditions that may increase the likelihood of landslides, and where these conditions are concentrated across the landscape. Exposure is

defined as: the people and infrastructure that exist in the zones where landslides are likely to occur. Vulnerability is defined as: the demographic factors that may affect the hazard's impact on exposed areas/populations. Based on disaster risk literature, including the United Nations' Sendai Framework for Disaster Risk Reduction 2015-2030 (UNDRR, 2015), these three components combine to define total landslide risk (Ming et al., 2022). Our goal was to map where these factors overlap to reveal the zones with the highest risk of landslides causing physical and societal damage.

II. Hazard

We first analyze two key physical drivers of landslide hazard: slope and geology. We calculated the slope from the ⅓ arc-second resolution Digital Elevation Model (DEM) using ArcGIS Pro's Slope tool, set to output slope values in degrees. These slope values were then reclassified into five hazard levels following thresholds drawn from Campbell et al. (1975), the California Geological Survey (CGS Map Sheet 58), and the U.S. Geological Survey (USGS) report *Soil Slips, Debris Flows, and Rainstorms in the Santa Monica Mountains and Vicinity, Southern California* (See **Table 1**). The latter was especially emphasized due to its direct relevance to our regional study area. This literature provided regionally specific thresholds that reflect the relative hazard of landslides based on slope steepness and known failure patterns.

To calculate hazard by geology, we obtained detailed geologic data from the National Park Service's (NPS) Geologic Resources Inventory (GRI) for the Santa Monica Mountains. We reclassified geologic units into landslide hazard levels using insights from the California Geological Survey, the NPS GRI, and the README documentation accompanying the dataset (See **Table 2)**. Landslide deposits (categorized as Quaternary 1 or Q1s) and artificial fill were classified as the highest hazard because they represent previously failed or human-placed

materials known to reactivate under stress. Unconsolidated Quaternary deposits, such as alluvium, were assigned moderate hazard due to their low shear strength and susceptibility to saturation. Older, lithified bedrock units were classified as low hazard due to their relative resistance to weathering. To apply these classifications, we calculated a new GeoRisk field in the attribute table using the Calculate Field tool and then converted the geology polygons to a raster with a 10-meter cell size using the Polygon to Raster tool. This ensured spatial alignment with the slope raster and allowed for combined hazard analysis in later stages.

Vegetation's impact on landslide hazard was incorporated using two datasets: land cover based on the USGS National Land Cover Dataset (NLCD) and Normalized Difference Vegetation Index (NDVI). With both being Landsat-derived raster layers, the raw data had a 30x30m resolution. Similarly to the geologic hazard layer, both of these vegetation hazard layers were resampled to a 10x10m resolution to be combined with other layers. These two layers were chosen to capture the effect of vegetation types and health, respectively, on landslide hazard susceptibility, as vegetation cover is often identified as a major mitigating factor for landslide occurrence (Shu et al., 2019; Pacheco Quevedo et al., 2022).

Essentially, vegetation is understood to stabilize slopes from erosion and landslides, with more robust vegetation cover types, such as forests and shrubs, to stabilize slopes more effectively than thin (grass) or bare (scree) vegetation cover. These differences in vegetation types were accounted for in the NLCD layer and then reclassified to a hazard score from 1 to 4 based on the classifications used in the heuristic method of Shu et al., 2019, summarized in **Table 3**.

Similarly, widespread vegetation mortality can increase landslide hazard, for which NDVI was incorporated (Li and Duan, 2023). NDVI was calculated using Landsat 8 bands 4 and 5 using Equation 1 (USGS, 2024b):

Eq 1: NDVI = (Band 5 – Band 4) / (Band 5 + Band 4)

Ranging from -1 to 1, a high NDVI value can be associated with strong, healthy, thick vegetation cover that is likely to stabilize a slope, while a low or negative value represents the opposite. Given the already numeric nature of NDVI, we did not need to resample the NDVI to a numerical index at this point, though it was normalized before being combined with the other hazard layers. While drought and other vegetation stressors could affect NDVI and landslide hazard, we were primarily interested in using NDVI to capture post-fire vegetation loss due to the January 2025 Palisades Fire. For this reason, two NDVI layers (pre-fire from August 2024, post-fire from December 2025) were used to produce two parallel landslide hazard maps. The wildfire vegetation transition is reflected in the NDVI layers with a dramatic decrease in NDVI, as areas within the Palisades burn area transition from landscapes with full vegetation cover to barren landscapes. This could not be represented solely using the NLCD layer, as the most recent NLCD layer available is from 2023.

Another critical factor influencing landslide occurrence is precipitation, which directly affects soil moisture and pore water pressure and can destabilize slopes. In particular, orographic precipitation tends to concentrate rainfall on windward slopes, increasing the water infiltration and contributing to slope failure in these areas. Incorporating a streamflow network derived from a filled DEM (using the Fill tool) and watershed analysis further enhances the understanding of hydrological pathways and water accumulation zones, which are essential for predicting areas prone to landslides (Montgomery and Dietrich, 1994). The streamflow network helps identify

drainage patterns and flow convergence zones where surface water accumulation can increase soil saturation, thereby elevating landslide susceptibility (Montgomery, 1994). Combining precipitation patterns with hydrological modeling provides a more comprehensive assessment of landslide hazards by integrating both the intensity and movement of water in the landscape.

The streamflow layer for this report was derived from the same USGS ⅓ arc-second DEM used in the slope analysis. To create a useful streamflow index based on our generated streamflow network, we created buffers with widths corresponding to 10 meters per stream order. With stream orders in our analysis ranging from 1 to 7, this meant that our stream buffers could be as wide as 70 meters, which seemed appropriate for the width of the major streambeds based on visual inspection of stream beds on ESRI satellite basemaps. This new stream order buffer feature class was rasterized, with each grid cell gaining the stream order of its stream vector as well, to account for the greater water flow in major streams. Additionally, 1981-2024 annual mean precipitation data from Oregon State University's PRISM Group was utilized as a gridded precipitation layer. This was taken at an 800 meter spatial resolution and resampled to a 10 meter resolution as with the other hazard layers.

III. Combined Hazard

To synthesize our identified contributors to landslide hazard, we used the raster calculator tool in ArcGIS Pro. In the absence of an explicit literature-based framework, we chose to weigh the three types of hazard – geologic hazard (slope, geology type), vegetation hazard (land cover, NDVI), and hydrological hazard (precipitation, streamflow) – equally in our composite hazard index (Equation 2). This meant that each hazard type had a maximum score of 1 and the composite hazard had a hypothetical maximum score of 3.

Eq 2: Hazard = (geologic hazard) + (vegetation hazard) + (hydrological hazard)

Within each of the three hazard types, we incorporated two data layers as previously discussed. Based on Campbell (1975), whose regional study in the Santa Monica Mountains emphasized slope as the primary control on soil slip susceptibility– while also recognizing the influence of geological materials on failure mechanics– we assigned our slope and geology layers 60% and 40% weights, respectively, in our hazard model. In the absence of other literature assigning explicit weights, we assigned the weights of the two layers in vegetation and hydrological hazard to 50%/50% in both cases. With the exception of NDVI, the data layers were normalized (divided) by their maximum values. In the case of NDVI, its highly non-uniform distribution meant that we had to take a different approach. We found that over 99% of the August 2024 NDVI values have a value of less than 0.33, so we took that to be the effective maximum value when normalizing the NDVI for both 2024 and 2025. The overall process, including weights, is summarized in Equation 3. For clarity, the values used to normalize are written in red, while the relative weights are written in blue.

Eq 3: $(0.4*Geology/5 + 0.6*Slope/5) + (0.5*LandCover/4 - 0.5*(3*NDVI)) + (0.5*Precipitation/700.427 + 0.5*Streamflow/7)$

This resulted in a composite index that ranged from roughly -0.096 to 1.689. This range was again normalized to an index of 0 to 5, summarized in Equation 4. Even though the coefficients used in Equation 4 were based off of the 2024 hazard score, it was applied to both the 2024 and 2025 composite hazard maps. This created the final indices used for the final composite hazard maps and incorporated into the overall landslide risk maps.

Eq 4: $New_Hazard = ((Old_Hazard + 0.096) / (1.689 + 0.096)) * 5$

IV. Exposure

To identify areas with heightened exposure to landslides, we created an exposure score based on land use and infrastructure data for the SMMNRA.

Exposure classification by land use was based on whether the area had dense physical infrastructure, such as institutions, commercial and residential buildings, which were given a high exposure risk classification compared to vacant land, such as agriculture, rural, and open land, which were given a low exposure risk classification (See **Table 4**).

As it is difficult to determine individual exposure risk for each land use type, the exposure classification for each land use was split into two classification scores: high or low risk. As a result, the exposure score was given equal intervals. Highly populated areas were classified as high risk from exposure in the Santa Monica Mountains. Open areas and mountain ranges were classified with lower risk, which is expected. Interestingly, half of the recreation area is classified as high risk, as the boundary has small residential neighborhoods dispersed across the mountain.

In addition to land use, infrastructure data were used to quantify areas of high human exposure to landslides. Roads, which connect even the most remote parts of the Santa Monica Mountains, are a critical component of the region's infrastructure (Zhou et. al., 2024). Electricity transmission and distribution lines, which power everyday life in the region, are also vulnerable to landslides. Both of these vital networks were considered when assessing exposure to landslides.

All roads are susceptible to being damaged or destroyed by landslides, but some roads are more important than others. Only a few major roads cross the SMMNRA from Highway 101 in the north to the Pacific Coast Highway in the south: Topanga Canyon Blvd (Highway 27), Las Virgenes Rd/Malibu Canyon Rd, Kanan Rd/Kanan Dume Rd, and Westlake Blvd/Mulholland

Highway/Decker Rd (Highway 23). If any of these four roads were affected by a landslide, large areas of the SMMNRA could be rendered inaccessible by car, and the diversion of traffic to other roads would cause significant delays. Fortunately, the U.S. Census Bureau's TIGER/Line shapefile for roads includes road type classifications, which identify county, state, interstate, and federal highways. Roads of these types were be prioritized over municipal and other, smaller roads when weighing the exposure score (**Table 5**).

In order to consider roads alongside raster data layers, the polylines had to be converted to a raster layer. Buffers were created using the Pairwise Buffer tool, then rasterized with the Polygon to Raster tool, based on road type: highways were given a 25-meter buffer, and all other roads were given a 10-meter buffer. These buffers represent the width of the road (given that a single lane is roughly 3.3 meters wide) plus an additional road width, to account for potential damage caused by a landslide near to, but not on top of, a road. With this, a raster layer for road exposure was created.

In order to account for the possibility that a single landslide could affect multiple roads, road density was also considered. Areas with a higher density of roads have inherently more infrastructure exposure because a landslide anywhere in the area is more likely to affect one or multiple roads. Road density was determined with the Line Density tool, which measures the length of road within a given area (in this case, km/km^2). Road density scores are classified in **Table 6**.

Finally, electricity infrastructure was considered as a part of the overall exposure score. Electricity infrastructure falls into two broad categories: transmission and distribution. Transmission lines connect electricity generation resources to local substations, and distribution lines bring electricity from substations to consumers. Transmission lines generally carry much

higher voltages than distribution lines: even the relatively low-voltage transmission lines in the SMMNRA carry 66 kV, while distribution lines operate at 34.5 kV or less, down to 4.8 kV (LADWP, CEC). Differing landslide effects on transmission and distribution lines can be roughly approximated using voltage: the destruction of a higher-voltage line will cut off electricity to more people (Huang et. al., 2024).

The process of creating a raster electricity infrastructure exposure layer was similar to that used in the road infrastructure layer, with differing buffers created for different line sizes (buffer creation was informed by Southern California Edison's Right-of-Way guidelines). The final electricity exposure classifications can be found in **Table 7**.

All exposure values were normalized to a 0 to 5 scale before being combined with hazard and vulnerability scores.

V. Vulnerability

To evaluate demographic vulnerability to landslides across the Santa Monica Mountains National Recreation Area, a composite vulnerability score was created using six key indicators. These variables are widely used in disaster vulnerability research, representing groups that are more likely to experience challenges during hazard events like landslides, especially in preparation, evacuation, or recovery phases.

Initially, raw U.S. Census demographic tables were directly joined to TIGER/Line shapefiles by GEOID. However, this method quickly became inefficient due to the volume of fields and the strain it placed on ArcGIS Pro's performance. A more manageable solution was pivoted to by using a 2020 Census Block Group layer that already included many of the key demographic attributes needed, such as population age groups, household types, and total population counts.

For each indicator, the Field Calculator was used to create a standardized score ranging from 1 (least vulnerable) to 5 (most vulnerable). Breakpoints were defined based on the distribution of values across the study area and relevant literature. **Table 8** summarizes the factors used, their observed ranges, scoring methodology, and final weights in the composite vulnerability index.

Each score was calculated with a new field and expression in the attribute table. Then the following weighted formula was used to calculate the composite vulnerability index for each census block group:

Eq 5: Vuln_Score = (Elderly_Score * 0.25) + (Minor_Score * 0.15) + (Alone_Score * 0.15) + (Institutionalized_Score * 0.20) + (Density_Score * 0.15) + (Rural_Score * 0.10)

To prepare this vulnerability layer for integration with our hazard and exposure maps, the dataset was reprojected to WGS 1984 UTM Zone 11N, ensuring consistency across the project's spatial reference system. The Feature to Raster tool was then used to convert the final vector layer into a raster with a 10x10 meter cell size to match the resolution of the hazard layer.

The final vulnerability scores ranged from 2 to 5 across the Santa Monica Mountains area. While no block groups received a 1, this is expected based on the region's demographic characteristics and does not undermine the usefulness of the scale—it still identifies relative variation across space.

Certain census tracts possess more vulnerable demographics than others, but they can also be inaccessible to first responders. To predict regions physically vulnerable to a lack of first responder access, we created a network analysis on the service area of fire stations in the Santa Monica National Recreation Area. This network analysis indicated the time for a first responder to arrive anywhere in the SMMNRA.

Drawing from the County of Ventura & Los Angeles's GIS data hub, we gathered all the fire stations and performed a service analysis (Service Area) with the intervals of 5, 15, 30, and 60 minutes away from each facility (See **Table 9**).

The vulnerability score was calculated in equal intervals on a scale from 1 to 5. The highest service area duration was 60 minutes for dirt paths and mountainous ranges. It's important to note that the 45-minute range was rather minimal, as most hard-to-get-to ranges fall into the 60-minute range due to the lack of fire stations, with there only being eight on the premises.

VI. Integration

To synthesize the hazard, exposure, and vulnerability components into a unified landslide risk surface, we used the Raster Calculator tool in ArcGIS Pro. Each raster had a cell size of 10m and shared the same projection (WGS 1984 UTM Zone 11N). We applied equal weights to each layer to calculate total risk as: Risk = (Hazard + Exposure + Vulnerability) / 3. This provided a normalized score from 0 to 5.

Results

I. Hazard

The two hazard maps (**Maps 1-2**) depict landslide hazard scores across the Santa Monica Mountains. Our initial analyses were conducted in August 2024, followed by a supplemental post-fire analysis for February 2025. Hazard scores were derived from a combination of slope, geology, vegetation, and hydrology factors. Overall, hazard scores are highest in areas of steep, mountainous regions, while lower hazard values are generally concentrated in flatter coastal areas and valley bottoms. The February 2025 map (**Map 2**) reveals a notable increase in the extent and intensity of hazard scores, particularly within the Palisades burn perimeter. Across

both hazard maps, spatial patterns of elevated risk align with areas of steep, unstable, sparsely vegetated terrain.

II. Exposure

In our final exposure analysis, residential areas in the Santa Monica Recreation Area were classified as most at risk of exposure from a landslide, which was expected. In particular, exposure risk increased in proximity to main highways that extend across our study area, as seen in **Map 3**. This is anticipated as electricity and transport infrastructures are concentrated near 23, N9, N1, and Topanga highways, serving access to small canyon-side neighborhoods. For instance, **Map 4** indicates high exposure risk scattered in the center of the Santa Monica Mountains, with a concentration along the coast.

Combining both **Map 3** & **Map 4**, total risk from exposure remains concentrated in coastal areas of the Santa Monica Mountains as well as secluded neighborhoods in Calabasas.

III. Vulnerability

The demographic vulnerability map seen on **Map 6**, shows the risk of landslides based on six demographic indicators that would inhibit or affect a person's ability to evacuate when threatened by a natural disaster such as landslides. The highest scoring areas included around Malibu Point, Paradise Cove, and La Sierra Canyon. The area around Malibu point includes Pepperdine University which may contribute to the score of 5 for minors living in the area. This area also scored a 5 in population density and institutionalized contributing to the risk of the area. Both the Paradise cove and La Sierra Canyon areas scored 5s in the following categories – minors, institutionalized, population density, and rural areas leading to the high overall scores. It should also be noted that the area where the Palisades fire occurred had an overall demographic

vulnerability score of 4 due to population density, minors, rural population, and institutionalized scores.

In our final fire station network analysis (**Map 7**), readily accessible areas from nearby highways in the Santa Monica Mountains were classified as less vulnerable. These areas include most residential and institutional areas, which are generally within 15 minutes of a nearby fire station. Topanga and Point Mugu state parks were classified as the most vulnerable due to the lack of paved roads in the mountain canyons.

Map 8 displays the combined vulnerability scores for the study area, taking into account equal weights from the fire station network analysis and the demographic vulnerability analysis. This final map shows high vulnerability in areas like Topanga Canyon, where there is high population density and the roads and highways are rural and therefore difficult for emergency responders to access.

IV. Overall Risk

Map 9 illustrates the overall risk within the study area in 2024, taking into account hazard, exposure, and vulnerability at equal weights. This map shows the highest risk along PCH and the more densely populated and mountainous parts of Malibu. Additionally, **Map 10** gives us a similar combined map although with vegetation data in the hazard analysis in 2025, after the Palisades Fire. This change in vegetation data has greatly affected the hazard outcome as this area increased in risk from mostly 3s and 4s, to 4s and 5s. The Palisades Fire boundary is overlaid to visualize the burn area and its effects on landslide risk.

Discussion

Our final maps reveal clear spatial patterns in landslide risk across the Santa Monica Mountains, influenced by the combined effects of slope, geology, vegetation, hydrology, infrastructure, and demographics.

The hazard map, produced by integrating slope, geology, vegetation, NDVI, and hydrology, indicates the highest hazard zones. High-hazard areas align with steep terrain, as steeper slopes increase the gravitational force acting downslope, making slope failure more likely. Our hazard maps also correspond with areas underlain by unstable geologic units, suggesting that unconsolidated or weak materials have low shear strength and are more prone to erosion and collapse. Low vegetation cover significantly influenced the spatial distribution of high-hazard zones. The post-fire NDVI layer notably amplified hazard in the area affected by the Palisades fire, demonstrating the significant impact that wildfire-driven vegetation loss can have on slope stability. Our results also suggest that hydrological factors further increase landslide hazard by increasing soil saturation and pore water pressure—conditions that can initiate slope failure, especially when combined with steep and unstable terrain. Overall, our results indicate that landslide hazard is highest where multiple contributing factors converge, emphasizing the importance of a multi-criteria approach to terrain hazard assessment.

The exposure map, produced by integrating land use and infrastructure data, shows where humans are most exposed to landslide risk. As might be expected, the most densely populated regions of the study area – including much of Malibu, parts of Calabasas, and Brentwood – are most exposed to landslide risk. Where there are people, there is development, and development can be damaged or destroyed by landslides. The land use data layer, in conjunction with the road density layer, also identifies smaller pockets within the interior of the mountains which are also

more developed, and therefore more exposed. Road and electricity infrastructure layers, because of their linear nature, do not have a significant effect on the overall appearance of the exposure map. However, they do affect exposure ratings for individual cells, and the cells with the highest overall exposure are found where roads, electricity lines, and densely populated areas all overlap.

The vulnerability map, produced by integrating fire station service area and demographic data layers, depicts the variance in vulnerability to landslides and their associated damages within the study area. A network analysis of fire station service areas accounts for differing vulnerability during the minutes directly following a landslide event: areas closest to fire stations are understandably less vulnerable, as first responders can arrive more promptly to limit damage and tend to injuries. The inclusion of demographic data allows for a more nuanced look at human response to landslides, considering which groups (the elderly, the young, the institutionalized, etc.) might have a harder time evacuating during a landslide or struggle more in the aftermath. The final vulnerability map is notable for its intense variation between neighboring regions, unlike the exposure map, which shows more gradual trends. Since demographic information is compiled on a census tract scale, these census-designated blocks appear as distinct entities, when the people living on their borders might in fact be very similar to their neighbors and real demographic boundaries might be more blurred. Nonetheless, the vulnerability ratings represent average scores across each respective region and are the most accurate way of quantifying varying demographic factors that may affect landslide vulnerability.

Implementing the hazard-exposure-vulnerability framework, we combined the previous results to identify key patterns in landslide risk across the Santa Monica Mountains. When all three dimensions were combined in our final map (see **Maps 9 and 10**), clear hotspots emerged in areas where steep, unvegetated terrain overlapped with communities containing vulnerable

populations and limited access to critical services. Particularly at-risk zones include parts of the Palisades, Malibu, and other scattered communities. These findings underscore the multifaceted nature of landslide risk: it is not solely about where landslides are likely to occur, but also about which populations are exposed and how resilient their communities are in the face of disaster. While many high-hazard zones are expected in steep, fire-affected terrain, this study can be used to identify underserved or isolated communities that may face compounded barriers in disaster response and recovery. By integrating these three dimensions, our analysis not only identifies where landslides are most likely, but also who is most at risk. This approach can better inform targeted mitigation efforts in the most vulnerable areas. Thus, our results reinforce the importance of integrating both physical and social factors in hazard mapping.

While our landslide risk assessment provides meaningful insight into spatial risk patterns across the Santa Monica Mountains, we acknowledge several limitations that affect the scope and precision of our results.

First, the analysis we conducted was static, where we treated all of our input datasets such as slope and land cover, as fixed snapshots in time. Landslide risk is dynamic, and current risk assessments can be significantly improved by incorporating temporal inputs (Vodacek, 2021).

Second, our hazard classification is based on simplified scoring thresholds drawn from our literature review. While these thresholds are grounded in well-cited sources, they generalize complex physical processes and cannot fully represent real-world systems. At most, our analyses provide a broad spatial approximation of landslide risk, useful for identifying general patterns.

Moreover, the representation of exposure and vulnerability is limited by the resolution and availability of social and infrastructure data. Census-based data offer only a coarse proxy for

social vulnerability, and while useful, they cannot capture individual household conditions or real-time population changes. Similarly, our exposure layers include the spatial distribution of infrastructure but do not account for the condition, age, or resilience of these systems in times of disaster. Additionally, the vulnerability scoring system relies on classification schemes and weighting decisions that are inherently subjective. Although informed by literature and data distribution, these choices still involve researcher judgment and could be adjusted in different contexts or with input from experts.

To build on this work, future analyses could incorporate additional risk factors– such as rainfall intensity– to better reflect the complex interplay of physical conditions that contribute to landslides. Incorporating spatial and temporal datasets, including time-series vegetation indices or precipitation patterns, would allow for a more dynamic assessment of risk. Furthermore, using historical landslide occurrence data in combination with statistical modeling could help identify high-risk areas based on past patterns. Comparing those results with our current findings would provide a valuable method of validating the accuracy of our approach.

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